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Deterministic Origin of Symmetry in Confined Water Surfaces: A Structural Approach Based on Causal Recording of Local Rules

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Abstract

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Deterministic Origin of Symmetry in Confined Water Surfaces: A Structural Approach Based on Causal Recording of Local Rules

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Abstract

This study examines how water surface ripples evolve into spatiotemporally symmetric patterns under spatial confinement and sustained energy input. To describe this transition, this research adopt a pattern-oriented, crystallographic perspective. This research introduce physically grounded local rules that operate both internally and through boundary-driven interactions within discrete elements. Numerical simulations by this framework demonstrates that a causal recording framework can reconstruct the evolution of real water ripple patterns. In addition, validated against real-world water experiments, the results reveal that spatial confinement is a critical factor in transforming stochastic interactions into highly ordered Faraday wave-like patterns driven by local velocity heterogeneity. This research propose that this emergent regularity arises from a deterministic balance between spatial propagation limits and the superimposed interaction of regenerated forces. These findings suggest that complex fluid motion can be interpreted as an ordered registration of local force histories, providing a robust framework for understanding self-organization in confined physical systems.

Introduction

Patterns are ubiquitous in nature as the form of periodic shapes, color, movements, informational rules and rhythms. There patterns appear in a variety of contexts, ranging from living and non-living system, static and dynamic, to causal relationships and spatial order [1-5]. The ripple

and Faraday wave of water surface are well-known phenomenon of dynamic patterns. The dynamic behavior of these patterns has been successfully described and interpreted through the lens of wave mechanics, governed by the three-dimensional Laplace equation for velocity potential and Faraday instability [6-8]. Furthermore, recent advancements have demonstrated that these governing equations can be effectively leveraged to implement precise control technologies, such as through the use of artificial structures and metamaterials [9].

Crystals are well-ordered, repeated structures by atoms, molecules, and ions, another well known types of patterns. From a general global perspective, thermodynamic stability of a crystalline phases is governed to minimize its total Gibbs free energy including bulk and surface parts [10-12]. From another perspective, the crystal structure patterns can be interpreted as a collective behavior of the smallest units having its own rules such as molecules. The basic rules of molecules, such as atomic radius, molecular geometry, molecular conformation, coordination number, bond length, and local molecular interactions by secondary bonds such as hydrogen bonds, play a very important role in determining the overall crystal phase [13, 14]. At ambient conditions, liquid water is best described as a continuous network of H_2O molecules, where hydrogen bonds constantly distort, break, and reform [15, 16] If water ripples or Faraday waves are viewed as emergent patterns, they can be interpreted as collective behaviors governed by the local structural rules inherent in the molecular units.

Abstract cellular automata (CA) model consist of many simple identical components is capable of generating remarkably complex patterns [17]. The fundamental strength of the CA model lies in local interactions of local interactions according to predefined rules. Furthermore, the versatility of CA as a tool for dynamic modeling has been investigated across a wide range of fields for simulating complex systems in

biology, computer science, and particularly materials science [18].

The Lorenz system is widely recognized as a foundational model for explaining chaos and complex nonlinear dynamics [19]. While it is typically studied for its sensitivity to initial conditions, I focus on two specific attributes of this system. First is the inherent causality, which establishes a rigorous link between temporal states [20]. Second, I focus on the system's ability for autonomous evolution governed by internal rules and initial state. In this view, complex patterns can be emerged based solely on its own deterministic equations, without the need for continuous external intervention. Despite their success as theoretical frameworks, these two models face inherent challenges in directly modeling or describing actual natural phenomena.their abstraction often limits the integration of real-world physical constants and variables, such as mass, velocity, etc. which are essential for a comprehensive representation of physical world.

To bridge the gap between abstract deterministic models and physical reality, this work formulates a grounded deterministic framework where the evolution of water surface patterns is governed by local vector-based rules operating strictly under Newtonian dynamics and Hooke's law. Importantly, this approach treats the water surface as a structurally homogeneous medium. The model is constructed from three fundamental principles: causal force propagation, geometric recording of force history at the front, and local regeneration at interaction boundaries. Together, these principles define a field that records time-dependent mechanical influence, allowing complex surface behavior to emerge. The following sections introduce the structure of this framework and its governing equation, before examining its implications through numerical simulations.

Conceptual origin of deterministic Water Surface Pattern Formation

In this work, I attempt to find the fundamental cause of water surface pattern formation in the spatio-temporal recording of local inputs, motivated by the ‘Walking droplets’ experiment. Successive collisions of liquid droplets generate repetitive ripples on the surface, and the droplets move forward by interacting with the wave structures by their past inputs [21]. It can be interpreted that continuous structures emerge from an initial stimulus as a historical recording.

In this system, the observed water surface response is decomposed into three sequential mechanisms: the generation of a source force, the temporal history of force transfer governed by a causal horizon, and the regeneration of the source triggered by the obstruction of front propagation at spatial boundaries (Figure. 1). The collective surface behavior emerges from the time-ordered execution of these rule-based local mechanisms. Consequently, the apparent propagation of water surface patterns during propagation reflects a recorded sequence of local force states.

Source force refers to a localized mechanical input applied to the water surface, acting on a discrete area to establish a clear force direction. Motivated by the periodic impacts in ‘walking droplets’ experiments, this initial stimulus triggers a deterministic response within the water surface. Upon application, the surface element undergoes a mechanical displacement, and a restoring response arises due to elastic and gravitational effects inherent to the fluid interface. This interaction sets the fundamental scale and orientation for subsequent pattern formation. The response of the local site is governed by predefined mechanical rules, such as inherent stiffness and mass, which effectively encode the input into a historical state. In this context,

the source force serves as the geometric and causal origin providing the temporal recording of the complex water surface patterns emerge through autonomous evolution.

Force transfer describes how the mechanical influence of a localized source force manifests at neighboring surface regions over time. Following the initial input, adjacent areas of the water surface exhibit delayed mechanical responses governed by a finite propagation speed. This phenomenon defines the system's causal horizon. Importantly, this process does not involve the continuous transport of a physical entity; it reflects a sequential recording of local force states. Directionally encoded force vector is systematically transformed into kinetic responses at each site as the causal front passes. Each water surface element begins its own autonomous evolution by the identical deterministic rules as the source, ensuring the causal continuity of the emergent water surface pattern without sustained external intervention.

Force regeneration indicates the re-emergence of surface motion following mechanical interaction with spatial boundaries or geometric constraints. It denotes the continued execution of the original rule-based sequence under restricted conditions. When the propagating front reaches the edge of this defined functional domain, the spatial propagation is obstructed, and the local state is reconfigured. At this interface, the incoming kinetic response is deterministically converted into a downward directed force. This stored state then serves as a localized causal trigger, allowing the water surface pattern to persist and evolve autonomously.

A local deterministic causal framework of surface records

The mathematical framework employed here emerged through a gradual refinement of a

previously proposed description of force-mediated surface propagation [22]. In that work, an eikonal-type formulation was used as a temporal registration device, encoding when a given spatial location becomes mechanically eligible to respond to a localized force input [23].The present study builds directly on that conceptual foundation, while extending it to a fully deterministic force–mass framework applicable to real water surfaces. The eikonal front is retained not as a wave construct, but as a causal horizon that governs the temporal ordering of local mechanical responses. This allows the surface evolution to be described through a sequence of local force applications, transfers, and regeneration governed by Newtonian dynamics and Hooke's law.

The full governing equation introduced below is constructed to reflect the three mechanisms outlined in the system description: (i) localized source force generation, (ii) temporally ordered force transfer defined by a finite propagation rule, and (iii) deterministic regeneration through energy conversion within the local fluid mass. The governing equation is formulated to encode how a localized mechanical input becomes accessible across space through time and is locally executed by the surface element.

$$\Psi\left(r,t\right)=A\alpha\left(\tau\right)H\left(\phi\right)$$

In this equation, $\Psi\left(r,t\right)$ represents the recorded mechanical state of the water surface at position r and time t. Importantly, ψ does not denote a transported physical quantity, but the outcome of a locally executed force–mass interaction conditioned by the accessibility of prior force history.

Source Force Term A

The coefficient A represents the magnitude and spatial distribution of a localized mechanical

input applied to the water surface at position r_s. In this framework, the source force is defined by the interaction between the input magnitude and a structural domain, expressed as:

$$A = A \times \beta$$

The introduction of the β field defines the boundaries within which the local deterministic rules can be executed, effectively constraining the propagation of force to a predefined structural area. When the initial force is applied, it acts on a local surface element and establishes a well-defined initial direction for the surface displacement. This interaction sets the characteristic scale and geometric orientation for all subsequent motion. The evolution of the pattern emerges entirely from the deterministic mechanical response of each discrete site within the permitted spatial domain.

Local Mechanical Response $\alpha(\tau)$

The function $\alpha(\tau)$ describes the time evolution of the local surface response following force application:

$$\alpha(\tau) = e^{-\gamma\tau}\cos(\omega\tau), \qquad \text{with} \quad m \cdot \ddot{\alpha} + c \cdot \dot{\alpha} + k \cdot \alpha = 0$$

This term corresponds to the solution of a damped harmonic response governed by the effective mass m, damping coefficient c, and restoring stiffness k of the water surface. Physically, it represents how a displaced surface element undergoes restoring motion due to gravitational and elastic effects. The response is entirely local and deterministic, and its temporal evolution is recorded rather than propagated.

Causal Horizon Operator $H(\phi)$

The term $H(\phi)$ defines the spatiotemporal accessibility of the recorded force history:

$$H(\phi) = \Theta\left(t - \frac{|r - rs|}{V} \right)$$

where V is the finite propagation speed governing the expansion of causal accessibility. This operator determines whether a given surface location can access the force history up to time τ. As a result, different spatial locations execute identical mechanical rules over different accessible time intervals. Within this framework, the apparent propagation of water surface patterns arises from the spatial expansion of temporally accessible force histories. Each surface element responds by executing the same local force–mass relation once the causal condition is satisfied. When the accessible force history reaches a geometrical boundary, the boundary initiates a new localized source force, restarting the same deterministic process under altered geometric constraints. This regeneration mechanism preserves the directionality of the force–mass interaction while allowing continued surface propagation. The integration of the Heaviside function $H(\phi)$acts as a causal switch that ensures a continuous temporal flow via a constant velocity V. This invariant speed allows the system to preserve the original force history across the surface without distortion, providing as a consistent causal recording process.

Numerical realization of confined water surface records under local deterministic causality

Figure. S1(A) and movies S1A present the numerical simulation of single surface evolution generated solely by the governing equation of local element, without imposing any boundary-induced regeneration or external constraints. A single localized source force is applied at the center of the domain, and the subsequent surface response is computed according to the deterministic and causal accessibility defined in the model. The upper panels show the two-dimensional surface record at increasing times. As time progresses, a radially expanding surface

pattern emerges from the source location. This expansion arises from the sequential activation of local surface elements as their accessible force history becomes available. The spatial extent of the pattern increases monotonically with time, that reflect the finite speed at which causal accessibility expands across the surface. The lower panels display one-dimensional cross-sectional profiles taken through the center of the domain. These profiles reveal a structured alternation in the recorded surface state as a function of position. In addition, each surface element executes a time-dependent mechanical response once activated. Notably, the spatial repetition observed in the cross-sections is not preassigned but emerges from the combination of the local restoring force vector and the temporal ordering imposed by the predefined rules of the initial source. A characteristic feature of the simulation is the asymmetric structure at the leading edge of the expanding surface record. The front exhibits a truncated or partially developed structure. This behavior reflects that surface elements near the advancing front access only a limited portion of the force history and simultaneously experience a spatially distributed reactive force arising from local mechanical recoil. As a result, the local response near the front remains geometrically and temporally incomplete compared to regions closer to the source.

Figure. S1(B) and movies S1B examine how variations in the initial mechanical conditions influence the resulting surface pattern record. Compared to Figure S1(A), the input force is increased, the effective mass is reduced, and finite damping is introduced. These changes localize the initially perturbed surface region while enhancing the directional magnitude of the response. Under these conditions, the leading surface record exhibits a pronounced directional amplitude aligned with the initial force orientation, whereas subsequent surface responses are strongly decreased. This contrast indicates that the earliest activated surface elements retain a dominant portion of the force history, while later responses are suppressed due to geometric dispersion and damping. The cross-sectional profiles emphasize that the spatial distribution of recorded surface states is highly sensitive to the initial force–mass configuration. Small changes

in the applied force and local inertia can produce qualitatively different surface evolution patterns, even under identical governing rules. This result highlights that surface propagation in the present framework depends on the mechanical conditions under which the source force is applied.

Figure 2(A) and movies 2A show the surface record produced by a single localized source force under finite damping. Figure 2(B) and movies 2B display the surface records generated by two localized source forces, identical compared to the source of Figure 2(A). The resulting patterns exhibit spatial regions where local mechanical responses are enhanced, as well as regions where they are locally diminished. These features arise without the introduction of additional mathematical rules or coordination mechanisms and follow directly from the interaction of independently executed local force histories. It is shown that compatible directional tendencies of accessible force records reinforced the local response. In contrast, the coexistence of opposing directional tendencies reduced the local force vector. This behavior indicates that the mechanical record retains directional information and that local force executions can coexist and interact in space. Consequently, the resulting pattern exhibits structural features analogous to wave interference in fluid dynamics. This emergent behavior demonstrates that complex superposition-like effects can be reproduced through the local interaction of independent force histories, confirming that the deterministic causal framework inherently accounts for the reinforcing and canceling behaviors characteristic of water surface interference.

Figure S2 and movies S2 validates the functional role of the β field as a spatial partitioning operator that restricts causal propagation. The governing rules are focusing on whether the β field successfully terminates the expansion of the surface record at predefined limits. The side-view profiles clearly demonstrate that the causal front is abruptly arrested upon encountering the β-boundary (0). This result confirms that the structural domain effectively determines the

spatial allowness of local surface elements.

Figure 3, movies 3A and movies 3B compare ripple-like surface records generated by the present formulation introducing regeneration force with experimentally observed water surface behavior under comparable geometric conditions. The regeneration mechanism was implemented via a coding algorithm designed to trigger a new source force at the moment the causal front makes contact with the spatial boundary where the β-field is defined as zero or where propagation is arrested. The regenerated force was assigned a magnitude equivalent to 50% of the input source force to account for energy dissipation during boundary interaction.

In panels (B) and (D), water droplets are introduced at the side and center of a confined container, respectively, and the resulting surface motion is monitored in real time. Panels (A), (C) show the corresponding numerical ripple-like surface records produced by a single localized source in geometrically similar domains, without introducing additional boundary rules. The simulated ripple surface exhibit spatial and temporal patterns that closely resemble those observed in the experiments, constrained by the defined β field. The geometric organization of surface structures imposed by the container boundaries is reflected in both the experimental images and the recorded force patterns generated by the model. This correspondence indicates that the present formulation captures key mechanical features governing water surface behavior through local force execution and temporal ordering alone. The observed correspondence suggests that describing water surface motion as a time-recorded execution of local mechanical interactions provides a natural and effective framework for interpreting experimentally observed surface patterns. Across both experiment and simulation, the governing formulation is unchanged. The only distinction lies in the spatial placement of the initial localized event, from which all subsequent surface patterns emerge through time.

Building upon the model's success in reproducing real water surface behavior, surface evolution

under highly restricted spatial accessibility is explored. In this context, the emergence of regular surface structures, resembling Faraday-like waves, is driven by two key factors: stringent spatial confinement and the sustained presence of force inputs.To simulate the continuous vertical oscillation characteristic of Faraday waves, the damping coefficient in this numerical framework was fixed at zero. Figure 4 present the surface records generated by the present formulation when the spatial domain is geometrically confined. The two-dimensional surface records exhibit repeating geometric patterns, while the corresponding one-dimensional cross-sections reveal regular alternation in the recorded mechanical state (Figure 4(A) and moives 4A). While local-rule-based patterns often exhibit symmetry-breaking behavior in open domains, the introduction of geometric constraints forces the accumulation and interaction of directionally encoded force histories into a highly organized state. As a result, the recorded mechanical states transition from irregular fluctuations to well-defined, spatiotemporally symmetrical patterns.

Figures 4(B) and movies 4B demonstrate that the resulting surface patterns are highly sensitive to the initial mechanical rules. A subtle shift from 0.5 to 03 in stiffness (k) qualitatively alters the recorded geometric configuration, highlighting that the emergent order is strictly governed by the local deterministic parameters rather than being a mere reflection of the boundary shape. As the spatial domain expands in Figures 4(C), (D), and movies 4C, D, the surface records evolve into increasingly complex yet strictly symmetrical geometric patterns. Despite this growing global complexity, the central core consistently preserves the fundamental pattern observed in Figure 4(A) throughout the temporal evolution. This persistence suggests that the local mechanical record maintains its causal integrity at the origin, even as the extended domain allows for a denser accumulation and interaction of force histories. In Figures 4(E), (F), and movies 4E, F the system is simulated under a distributed force environment where sources are uniformly applied across the domain. Notably, Figure 4(E) shows that the resulting symmetrical

pattern remains nearly identical to Figure 4(D) regardless of the number or arrangement of sources, suggesting that the initial physical state and spatial constraints are more fundamental in shaping the surface behavior than the specific positioning of inputs. Furthermore, Figure 4(F) demonstrates that minimal parameters (the local mass, element density) change leads to the emergence of highly symmetrical crystalline-like patterns. This phenomena confirms that the deterministic force-mass framework can effectively capture such sophisticated physical transitions. The observation of symmetric pattern transitions through variations in m and k can be interpreted as a phenomenon analogous to the instabilities observed in physical experiments, where patterns evolve according to the liquid medium (density/viscosity) and the vertical driving frequency. Interestingly, the patterns observed in Figure 4(D) at t=750 and t=1250, as well as in Figure 4(F), exhibit a striking resemblance to the experimental Faraday wave patterns observed in a square vessel brimful of water, with the vertical reflection of a light beam, reported in previous literature [2]. Reported results are controlled by local constraints and parametric excitation, the structural transitions in this study can be interpreted as a result of introducing velocity heterogeneity into the propagation front.This suggests that the present simulation framework not only replicates the visual outcomes but also effectively encapsulates the underlying experimental conditions where physical parameters dictate the emergent symmetry.

Figure S3 and movies S3 investigate the influence of stochastic inputs on the emergent surface record by applying random spatial positioning and temporal onset to the sources. While these randomizations lead to a breakdown of spatial symmetry, the system remarkably maintains its temporal symmetry, with each local element continuing to execute its periodic mechanical response. This result confirms that while global spatial order is governed by the boundary-induced confinement and the fundamental temporal rhythm is an intrinsic property of the deterministic local rule. This results highlights that the spatial restriction of the domain remains

the dominant factor in ensuring the long-term structural organization of the surface record.

The collective results of Figure 4 illustrate a profound transition from stochastic surface fluctuations to highly ordered geometric structures, observed through a pattern-oriented perspective. While the local-rule-based surface record exhibits symmetry-breaking behavior in an open domain, the introduction of spatial confinement forces these independent mechanical histories to interact and synchronize, forming symmetrical patterns. This transition suggests that the emergence of global symmetry is not a result of pre-assigned coordination, but a deterministic outcome of the balance between spatial propagation limits and the superimposed interaction of regenerated forces. Ultimately, this framework also suggests a robust foundation for interpreting complex liquid medium patterns as the ordered execution of local interactions, revealing the intrinsic link between spatial restriction and the birth of symmetrical geometric order.

Discussion

The present results suggest that the emergence of spatiotemporal symmetry in water surface behavior reflects a structural reorganization of nonlinear ripple dynamics governed by the balance between internal propagation and boundary-mediated input under spatial constraint This study also proposes a new approach describing water surface behavior, transitioning from traditional dynamical systems to a pattern-oriented, crystallographic perspective. By introducing a physically grounded local rules that operate internally and occur externally within discrete element, it demonstrate that causal recording frameworks can successfully reconstruct real water surface structure evolution. This approach allows for a unified "grammar" that reproduces both simple ripples and complex, Faraday-wave-like symmetric patterns by controlled space area without the need for phenomenological corrections. A central finding of this work is that spatial confinement acts as a catalyst for order; it transforms the inherently nonlinear and

symmetry-breaking tendencies of local rules into spatiotemporally symmetrical patterns. I propose that this emergent regularity arises from a delicate balance between spatial propagation limits and the superimposed interaction of regenerated forces. Interestingly, The simulated patterns exhibit a striking resemblance to actual Faraday waves, where structural transitions mirror physical modulations driven by the medium, frequency, and velocity heterogeneity. This correlation demonstrates that variations in m and k effectively encapsulate the essential parametric dependencies of real-world water surfaces. The causal recording framework proves to be an intrinsically linked representation of fluid behavior under parametric excitation, reflecting actual experimental conditions. This deterministic causal recording framework suggests that what we often perceive as stochastic or chaotic fluid motion may be reconstruct an ordered registration of local force histories waiting to be structured by their environment. By revealing the intrinsic link between spatial restriction and the birth of symmetrical order, this study offers a robust pathway for interpreting self-organization in confined physical systems.

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Data and materials availability

All data are available in the main text or the supplementary materials. The source code is available at: https://github.com/junhyunglee73/real-space-water-surface-dynamics.git

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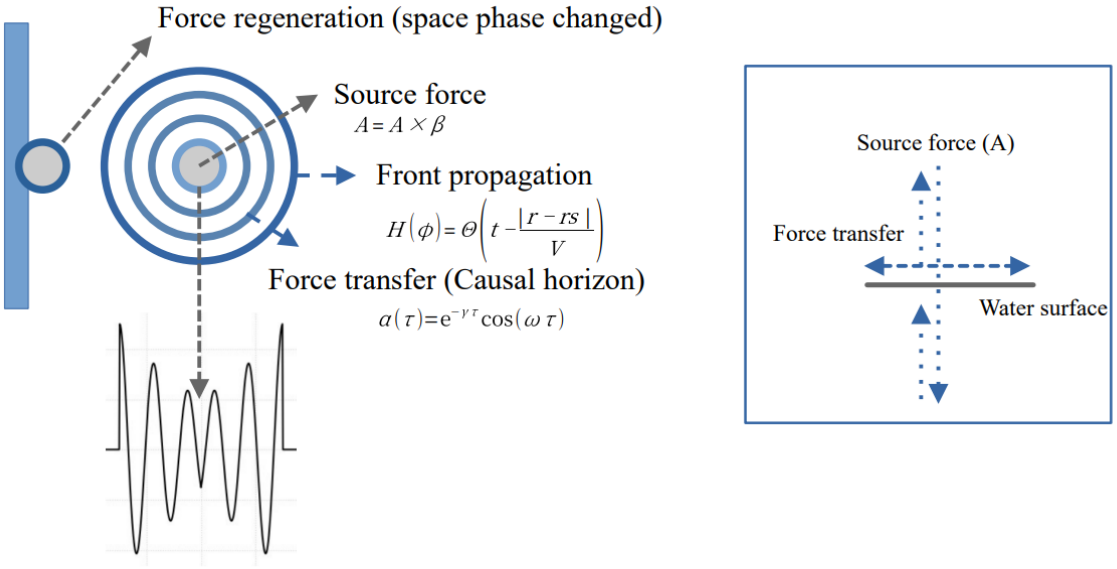


Figure. 1. Conceptual schematic of deterministic water surface pattern formation by causal recordings. The surface response is decomposed into three sequential mechanisms: (i) generation of a localized source force, (ii) force transfer constrained by a causal horizon that defines information accessibility in time and space, and (iii) force regeneration through deterministic energy conversion and directional reconfiguration.

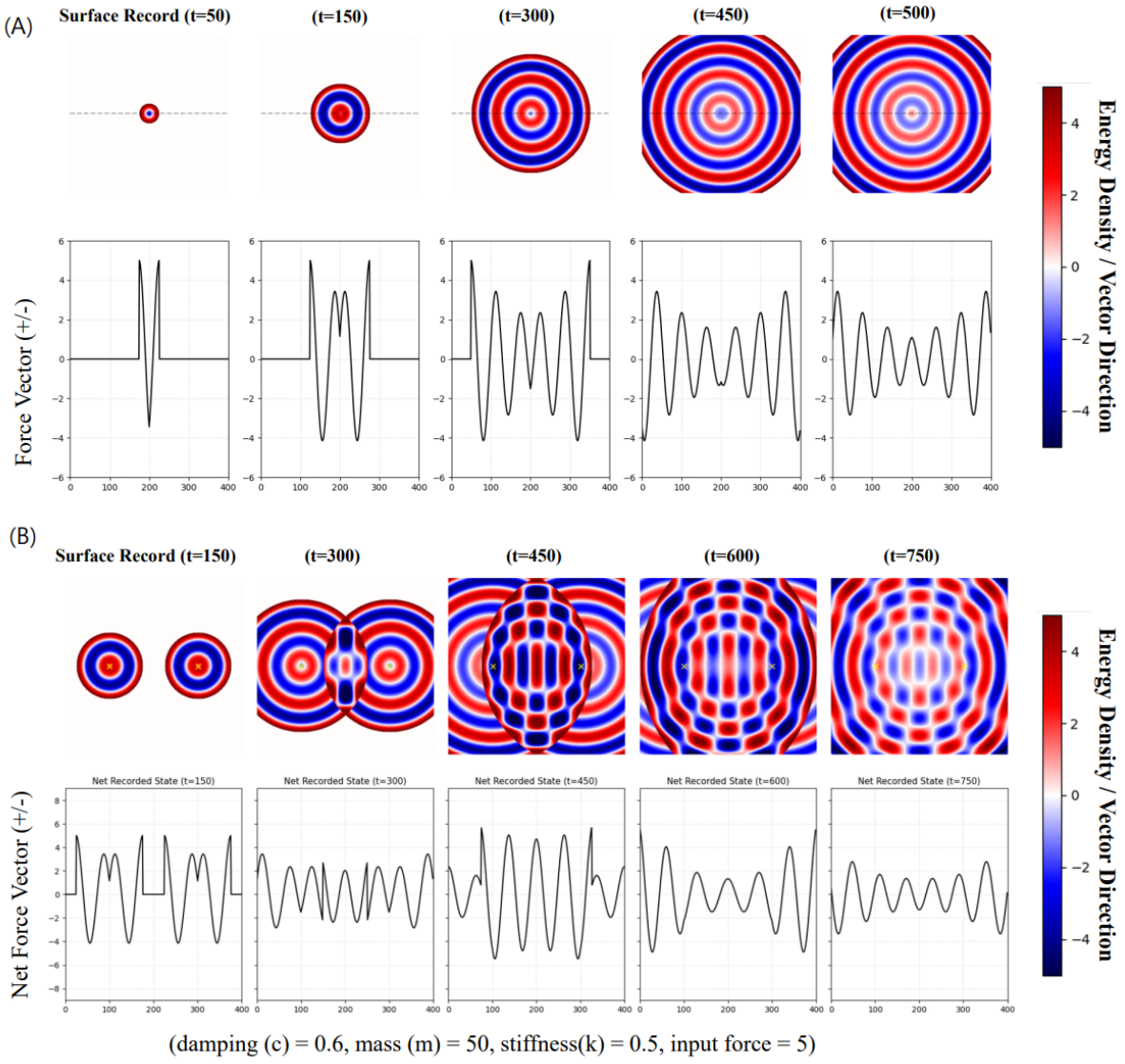


Figure 2. Emergent interference patterns from interacting causal sources. (A) Surface record of a single damped source. (B) Interaction of two identical sources where compatible force histories reinforce the local response, while opposing histories lead to local cancellation. The resulting pattern demonstrates that wave-like interference emerges deterministically from the spatial coexistence of independent local force executions.

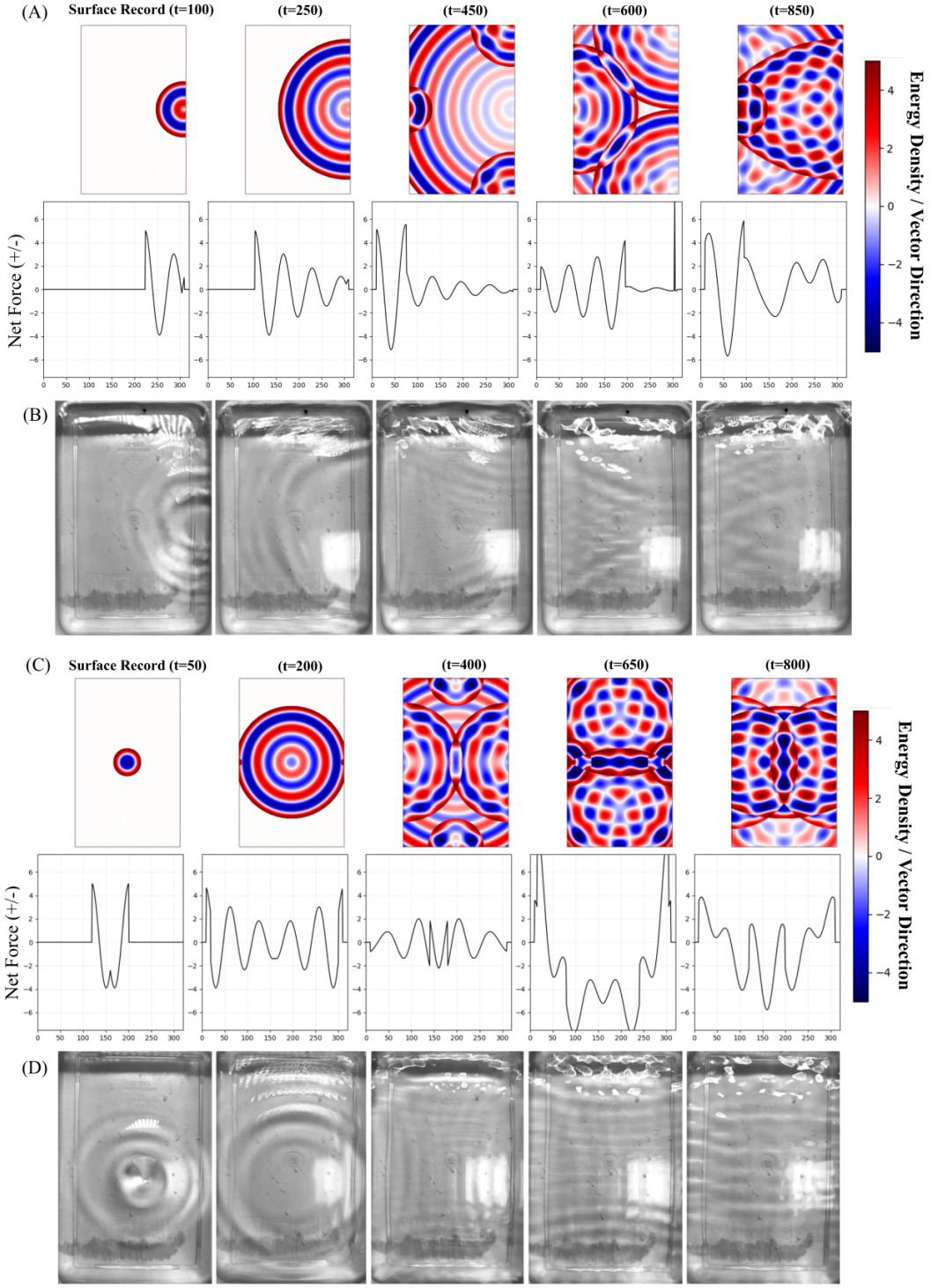


Figure. 3. Comparison between simulated surface records and experimental water surface pattern(A, C) Numerical surface records generated by a single localized source within a restricted β-field. (B,D) Corresponding experimental water surface patterns obtained by introducing droplets at analogous positions in a container of similar geometric proportions.

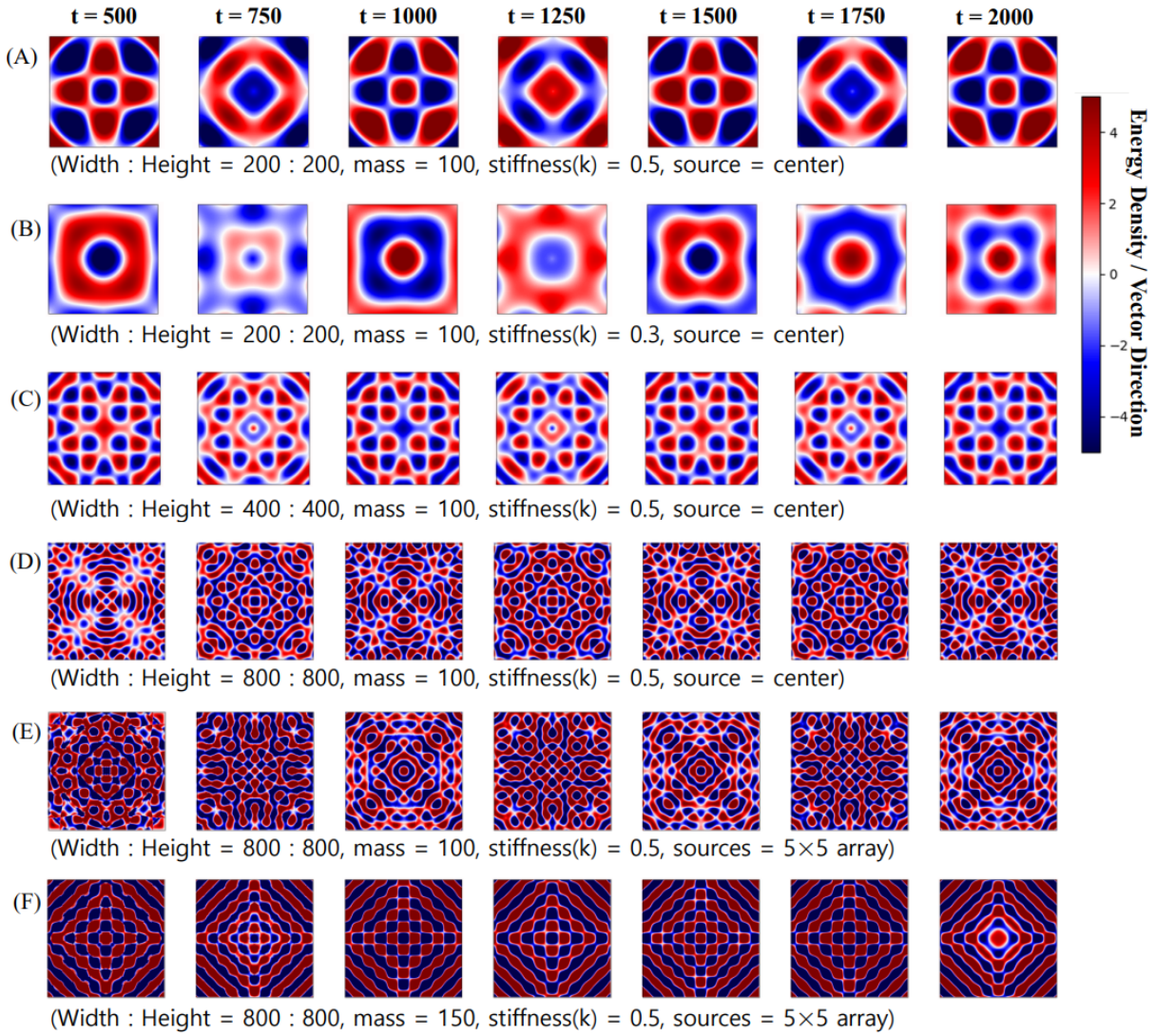


Figure 4. Emergence of symmetrical geometric order through constrained causal interactions. (A–B) Surface records where subtle shifts in local stiffness (k) qualitatively reconfigure the pattern density within a fixed β-field. (C–D) Expansion of the spatial domain showing increased global complexity while the central core preserves its fundamental structural integrity. (E–F) Collective responses under distributed inputs; (E) demonstrates that spatial constraints dominate the pattern over source positioning, while (F) shows mass and elements density driven transitions to crystalline similar structures. The resulting symmetry represents a deterministic balance between spatial propagation limits and the superimposed interaction of regenerated forces.

Supporting Information for

Deterministic Origin of Symmetry in Confined Water Surfaces: A Structural Approach Based on Causal Recording of Local Rules

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Experiment

Water surface patterns formation

Water was poured into a plastic container (19 × 12 cm), and surface patterns were generated by introducing droplets at locations corresponding to those used in the simulations. The resulting surface motion was recorded using the camera of an iPhone 14.

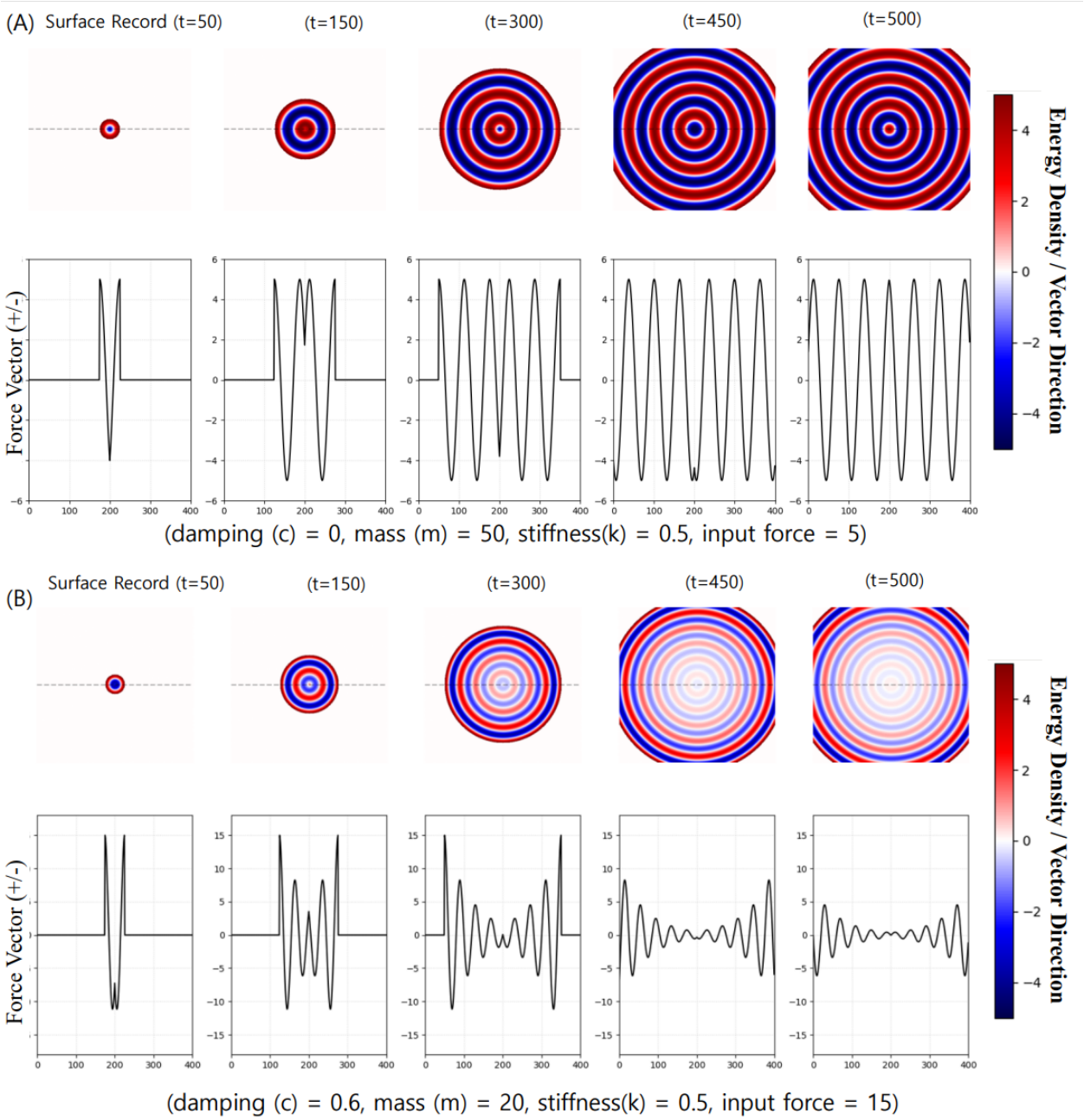


Figure S1. Emergent surface patterns generated by a single causal source in the absence of boundary regeneration. (A) Time-resolved snapshots of the recorded surface state produced by a localized source force governed solely by the causal record kernel. No boundary-induced source regeneration or additional spatial operators are applied. B) Increasing the source force and reducing the effective mass modifies the directional strength of the initial surface response, demonstrating that the propagation pattern is directly shaped by the input force parameters rather than imposed spatial constraints.

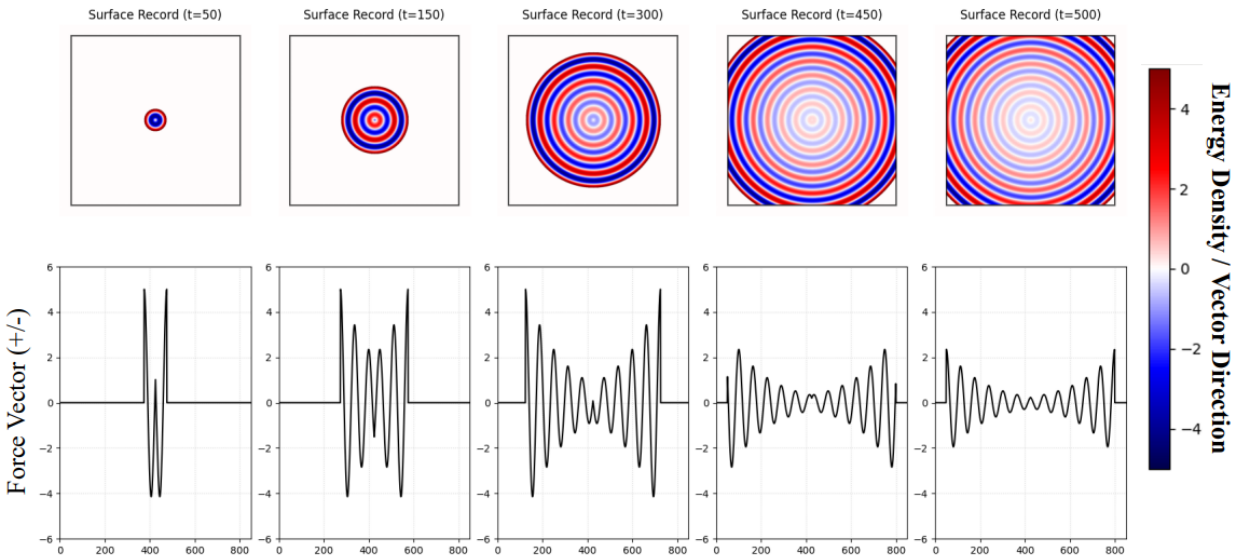


Figure S2. Validation of spatial propagation arrest via the β field. (Top) Sequential snapshots showing the radial expansion of the surface record being terminated at the spatial boundaries defined by the β field. (Bottom) Corresponding one-dimensional profiles illustrating the abrupt cessation of the causal front at the boundary.

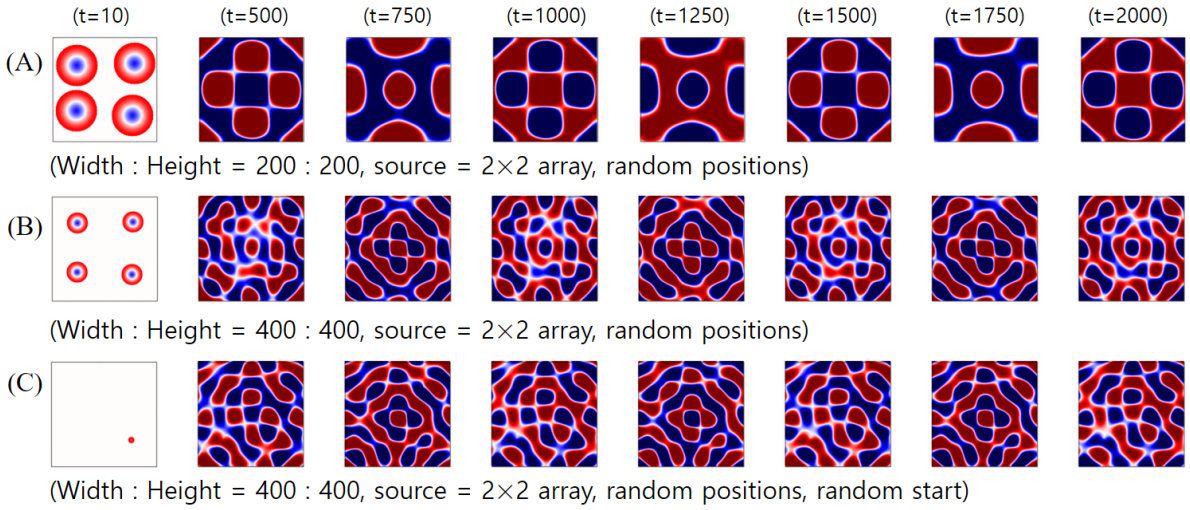


Figure S3. Robustness of temporal symmetry under stochastic spatial and temporal inputs. Sequential snapshots and profiles of surface records generated by sources with randomized spatial positions and onset times. Although the stochastic nature of the inputs leads to a breakdown of global spatial symmetry, the underlying temporal symmetry is maintained through the independent execution of local mechanical rules. This decoupling confirms that while the spatial domain βdominates the geometric organization of the pattern, the characteristic temporal rhythm remains an intrinsic property of the local deterministic framework.